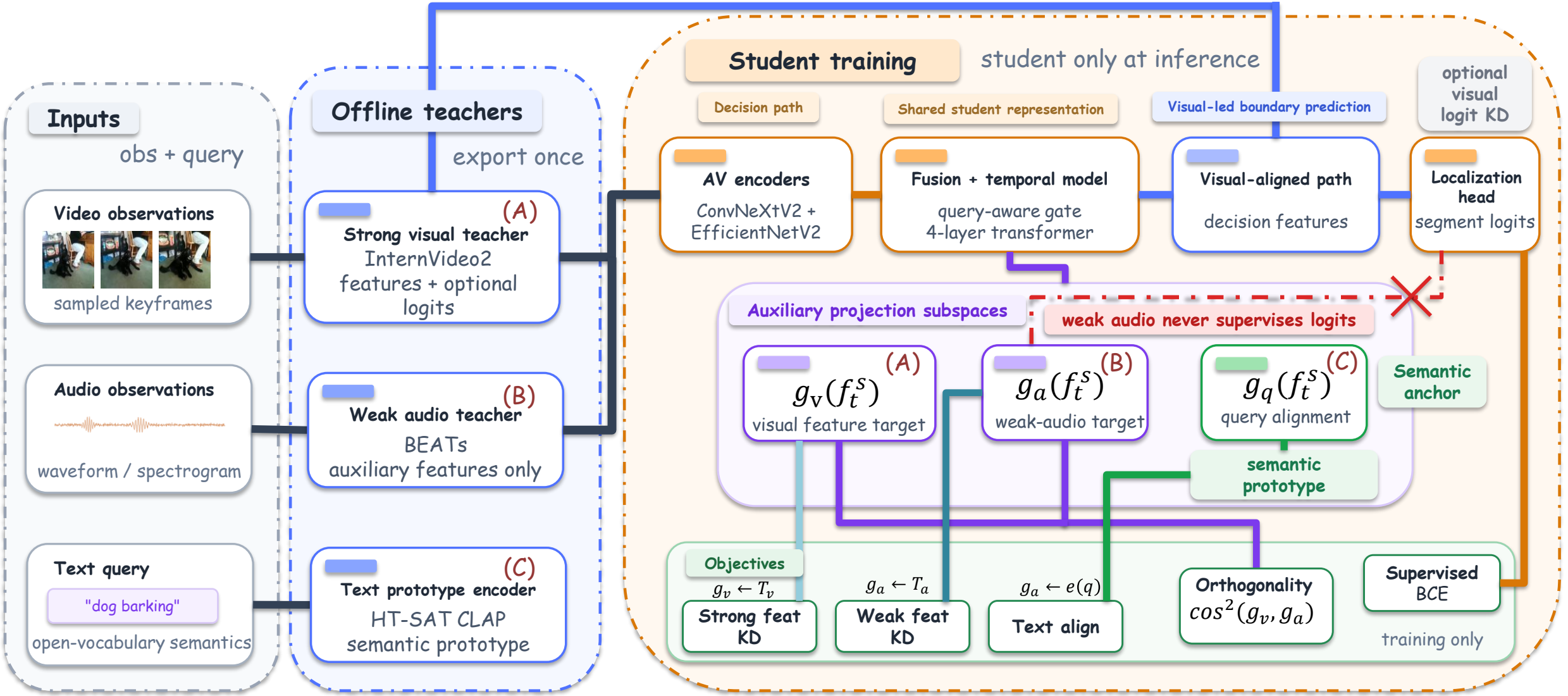

Inputs
obs + query
Video observations
sampled keyframes
Audio observations
waveform / spectrogram
Text query
"dog barking"
open-vocabulary semantics
Offline teachers
export once
(A)
Strong visual teacher
InternVideo2
features + optional logits
(B)
Weak audio teacher
BEATs
auxiliary features only
(C)
Text prototype encoder
HT-SAT CLAP
semantic prototype
Student training
student only at inference
Decision path
Shared student representation
Visual-led boundary prediction
optional visual logit KD
AV encoders
ConvNeXtV2 + EfficientNetV2
Fusion + temporal model
query-aware gate
4-layer transformer
Visual-aligned path
decision features
Localization head
segment logits
Auxiliary projection subspaces
weak audio never supervises logits
(A)
$g_v(f_t^s)$
visual feature target
(B)
$g_a(f_t^s)$
weak-audio target
(C)
$g_q(f_t^s)$
query alignment
Semantic anchor
semantic prototype
Objectives
$g_v \leftarrow T_v$
Strong feat KD
$g_a \leftarrow T_a$
Weak feat KD
$g_a \leftarrow e(q)$
Text align
Orthogonality
$cos^2(g_v, g_a)$
Supervised
BCE
training only